

Main theorem of Galois theory

Proposition: M/K Galois

$$K = M^{\text{Gal}(M/K)}$$

Corollary ①

$$\begin{array}{c} M \\ | \\ L \\ | \\ K \end{array} \leadsto L = M^{\text{Gal}(M/L)}$$

$$(2) \quad \left\{ \begin{array}{c} \text{subextensions} \\ L/K \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \text{subgroups} \\ H \leq \text{Gal}(M/K) \end{array} \right\}$$

$$L \longmapsto \text{Gal}(M/L)$$

is injective

(3) $\left\{ \begin{array}{c} \text{subextensions} \\ L/K \end{array} \right\}$ is a finite set

Proposition (2): M/K : Galois
 $H \leq \text{Gal}(M/K)$

$$|\text{Gal}(M/M^H)| \leq |H|$$

$$\Leftrightarrow [M:M^H] \leq |H|$$

Rmk: In fact, this will imply

$$H = \text{Gal}(M/M^H)$$

Pf: Suppose that $M = K(\alpha) \cong \frac{K[x]}{(m_\alpha(x))}$

Then M is also equal to $M^H(\alpha)$

$$\cong \frac{M^H[x]}{(m_{\alpha, M^H}(x))}$$

and $[M:M^H] = \deg m_{\alpha, M^H}(x)$

Consider $\underbrace{\{\sigma(\alpha) : \sigma \in H\}}_{S_\alpha}$: has at most $|H|$ elements

This should be the set of zeros of $m_{\alpha, M^H}(x)$

We are led to considering

$$f(x) = \prod_{\alpha \in S_\alpha} (x - \alpha) \in M[x]$$

This product is fixed by H

$$\Rightarrow f(x) \in M^H[x]$$

Moreover, $f(\alpha) = 0 \Rightarrow m_{\alpha, M^H}(x) \mid f(x)$

$$\Rightarrow [M : M^H] = \deg m_{\alpha, M^H}(x)$$

$$\leq \deg f(x)$$

$$= |S_\alpha|$$

$$\leq |H|.$$

This completes the proof of Prop (2)

When M is monogenic over K ,

Lemma: M/K is Galois

$\Rightarrow M/K$ is monogenic

Pf: By induction on $[M:K]$

$K \nexists \alpha \in M$

$|\{ [M:K(\alpha)] < [M:K] \}$

$K(\alpha)$

$|\{ [K(\alpha):K] > 1 \}$
 K

Inductive Hypothesis: $M = (K(\alpha))(\beta)$
 $= K(\alpha, \beta)$

Note: Can assume that M & K
are infinite fields & $\alpha, \beta \neq 0$.

$\Rightarrow \exists$ Infinitely many elements in M
of the form $\alpha + t\beta$, $t \in K$.

Consider the extensions $L_t = K(\alpha + t\beta)$

V/M

$$\Rightarrow \exists t \neq t' \in K \text{ s.t.}$$

Only finitely
many subextns
of M/K

$$L_t = L_{t'} = L$$

$\downarrow$ $\downarrow$
 $\alpha + t\beta$ $\alpha + t'\beta$

$$\Rightarrow \alpha + t\beta, \alpha + t'\beta \in L$$

$$\Rightarrow \underbrace{(t - t')\beta}_{\neq 0} \in L$$

$$\Rightarrow \beta = \underbrace{(t - t')^{-1}}_{\in K \subseteq L} \underbrace{(t - t')\beta}_{\in L}$$

$$\in L$$

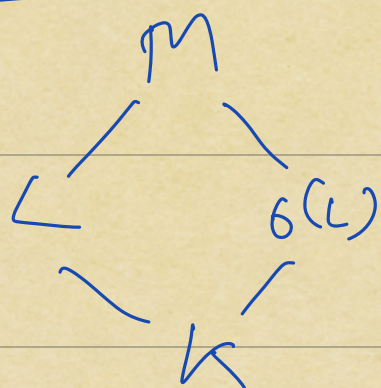
$$\Rightarrow \alpha = \underbrace{(\alpha + t\beta)}_{\in L} - \underbrace{t \cdot \beta}_{\in L} \in L$$

$$\Rightarrow K(\alpha, \beta) \subseteq L \subseteq M$$

$\parallel$
 M

$$\Rightarrow \left. \begin{array}{l} L = M \\ \parallel \\ K(\alpha + t\beta) \end{array} \right\} \text{monogenic.}$$

Proposition (3): $\sigma \in G, \sigma(L) \subseteq M$



$$\begin{aligned} (1) \quad \text{Gal}(M/\sigma(L)) &= \sigma \text{Gal}(M/L) \sigma^{-1} \\ &\subseteq \text{Gal}(M/K) \end{aligned}$$

(2) TFAE:

$$(a) \quad \text{Gal}(M/L) \trianglelefteq \text{Gal}(M/K)$$

normal

$$(b) \quad L = \sigma(L), \quad \forall \sigma \in \text{Gal}(M/K)$$

$$(c) \quad L/K \text{ is Galois}$$

(3) When (2) holds

$$\text{Gal}(M/K) / \text{Gal}(M/L)$$

$$\downarrow \cong$$

$$\text{Gal}(L/K)$$